

TASK 04 REPORT

Simulating an AI Interview Assistant using Prompt Engineering

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Task: Task 04 – Simulating an Assistant

1. Project Title

AI Interview Coach – A Prompt-Driven Technical Interview Assistant

2. Objective

The objective of this project was to design and develop an AI-powered Interview Assistant capable of conducting a structured mock technical interview. The assistant guides users through the interview process by collecting candidate information, asking role-specific technical questions, evaluating responses, providing constructive feedback, and generating personalized interview insights using prompt engineering techniques.

3. Project Overview

The AI Interview Coach simulates the behavior of a professional technical interviewer. Unlike a traditional chatbot that waits for user questions, the assistant actively drives the conversation by asking interview questions one at a time.

The interview process includes:

- Candidate onboarding
- Role selection
- Experience level selection
- Technical interview
- AI-generated evaluation
- Performance feedback
- Interview guidance

The assistant was built using **Botpress Cloud** and designed using structured prompt engineering techniques to create a natural, interviewer-led conversation.

4. Features Implemented

- Professional welcome message
- Candidate information collection
- Role-specific interview questions
- Experience-level based interaction
- One-question-at-a-time interview flow
- AI-generated answer evaluation
- Score out of 10 for every response
- Strengths and improvement suggestions
- Ideal answer generation
- Personalized interview tips
- Adaptive conversation flow
- Interview restart and termination commands

5. Prompt Engineering Techniques Used

The project incorporates several prompt engineering strategies:

Role Prompting

The AI is instructed to act as a professional technical interviewer instead of a general chatbot.

Structured Prompting

The interview follows a predefined sequence:

- Greeting
- Candidate Details
- Interview
- Evaluation
- Feedback

Instruction Prompting

Explicit instructions ensure the AI:

- asks one question at a time,

- evaluates responses,
- avoids revealing answers prematurely,
- maintains a professional tone.

Context Retention

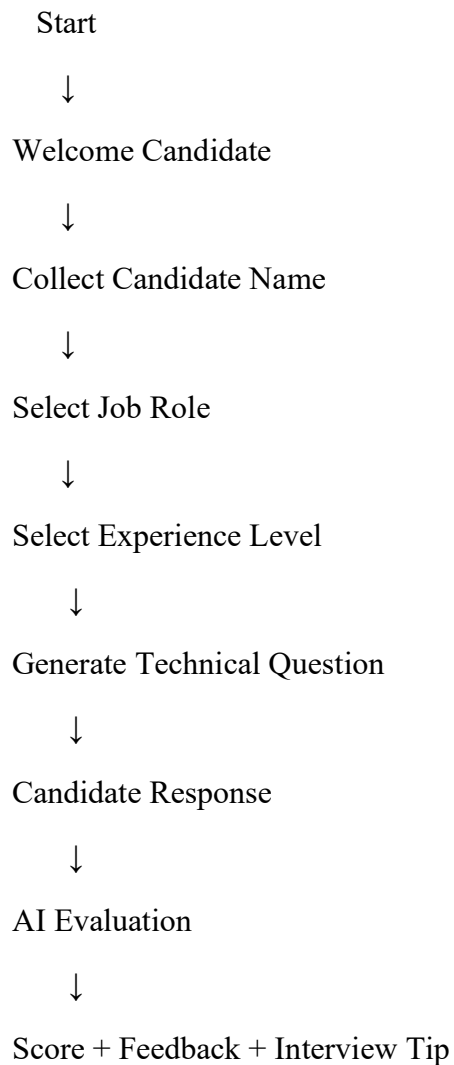
The assistant maintains interview context by remembering:

- Candidate Name
- Selected Role
- Experience Level

Adaptive Prompting

The difficulty of interview questions adjusts based on the candidate's previous performance.

Workflow





Next Question



Interview Completion

7. Sample Conversation

Assistant

Hello! Welcome to AI Interview Coach.

What is your name?

Candidate

Tanushri

Assistant

Which role are you preparing for today?

Python Developer

Candidate

Python Developer

Assistant

Question 1:

What is the difference between a List and a Tuple in Python?

Candidate

Lists are mutable whereas tuples are immutable.

Assistant

Score: 8/10

Strengths:

- Correct understanding of mutability.

Areas for Improvement:

- Mention syntax and practical use cases.

Ideal Answer:

Lists are mutable sequences defined using square brackets, whereas tuples are immutable sequences defined using parentheses. Tuples are generally faster and suitable for fixed data.

Interview Tip:

Support conceptual answers with examples whenever possible.

8. Challenges Faced

- Designing the assistant to proactively lead the conversation instead of waiting for user prompts.
- Maintaining interview context throughout the conversation.
- Structuring prompts to ensure one-question-at-a-time interaction.
- Ensuring feedback remained consistent and constructive.
- Improving prompt instructions to generate role-specific technical questions.

9. Learning Outcomes

Through this project, I gained practical experience in:

- Prompt Engineering
- Conversational AI Design
- AI Persona Development
- Structured Prompt Writing
- Role-based Prompting
- Multi-turn Conversation Design
- Context Management
- AI Response Optimization
- Workflow Design using Botpress

10. Technologies Used

- Botpress Cloud
- Large Language Models (LLMs)
- Prompt Engineering
- Conversational AI
- Workflow Automation

11. Future Enhancements

Future improvements planned for this project include:

- Resume Upload Integration
- Resume-based Interview Questions
- Downloadable Interview Report (PDF)
- Voice-based Interview Experience
- Performance Dashboard
- Interview History Tracking
- AI-generated Learning Roadmap
- Multi-role Interview Support
- Integration with Python Flask Backend

12. Conclusion

The AI Interview Coach successfully demonstrates how prompt engineering can be used to build an intelligent conversational assistant that simulates real-world technical interviews. The project showcases structured prompting, role-based conversation design, adaptive questioning, and AI-generated feedback, providing candidates with a practical interview preparation experience.